Juan David Muñoz Bolaños
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17. April 2026

Excelitas Technologies Göttingen GmbH
Königsallee 23
37081 Göttingen
Deutschland

Bewerbung als Systemingenieur optische Messtechnik | Ref. D37081

Sehr geehrte Damen und Herren,

für die Entwicklung hochkomplexer optomechanischer Module bei Excelitas Technologies in Göttingen bringe ich fundierte Expertise in der Prüfaufbau Entwicklung sowie der Metrologie und Systemvalidierung mit. Als promovierter Physiker Engineaur mit Schwerpunkt auf optischer Wellenfront Kontrolle und datenbasierter Prozesssteuerung bewerbe ich mich mit Interesse auf die Stelle als Systemingenieur optische Messtechnik (Ref. D37081).

Mein Hintergrund in der Entwicklung komplexer Messumgebungen passt ideal zu den Anforderungen Ihres Standortes als Kompetenzzentrum für Halbleiter Equipment. Während meiner Promotion habe ich vier Jahre lang präzise Prüfaufbauten von Grund auf entworfen, realisiert und qualifiziert. Ein Schwerpunkt lag auf der Messung und aktiven Korrektur von Wellenfrontaberrationen (ACS Photonics 2025), ein Prinzip, das direkt auf die Qualifizierung von Optik Modulen und Faserbaugruppen übertragbar ist. Ich bin versiert darin, Messbudgets zu kalkulieren, Systemschnittstellen zu definieren und detaillierte Sensitivitätsanalysen unter Berücksichtigung von Standards durchzuführen.

Die automatisierte Auswertung und Validierung von Messergebnissen ist ein Kernbestandteil meiner Arbeitsweise. Ich verfüge über fortgeschrittene Python Kenntnisse zur Erfassung und Analyse hochdimensionaler Messdaten. Am IPHT Jena habe ich das Paket SpectraMap entwickelt, um automatisierte Klassifizierungen und objektive Pass Fail Kriterien für Hyperspektral Data (Applied Sciences 2024). Diese Expertise ermöglicht es mir, Messumgebungen nicht nur aufzubauen, sondern auch deren Leistung systematisch im Sinne des Anforderungsmanagements zu verifizieren.

Meinen direkten Bezug zum Halbleitermarkt vertiefte ich während der ASML und ZEISS SMT Summer Schools, in denen ich Einblicke in die systemtechnischen Anforderungen moderner Lithographie und Inspektionssysteme gewann. Ich arbeite strukturiert, schätze interdisziplinäre Teamarbeit und bringe die notwendige analytische Tiefe für anspruchsvolle Systementwicklungen mit. Ursprünglich aus Kolumbien stammend, habe ich meinen Master in Jena absolviert und schließe meine Promotion in Innsbruck im Juni 2026 ab. Ich lebe seit vier Jahren in Österreich und freue mich darauf, meine Erfahrung in der industriellen Fertigung bei Excelitas einzubringen. Ich freue mich auf ein potenzielles Gespräch.

Mit freundlichen Grüßen,

Juan David Muñoz Bolaños
Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145
juan.munoz@i-med.ac.at | LinkedIn Profile

Summary

Physicist (Doctoral candidate) specializing in **adaptive optics (holography)** and **wavefront metrology**. Expertise in systems engineering (V-Model), **requirement management**, and development of complex measurement budgets according to the **Guide to the Expression of Uncertainty in Measurement (GUM)**. Upper Intermediate (B2) German and Fluent English.

Professional Experience

Jan'22 | Present — **Med. Univ. Innsbruck** — Innsbruck, Austria
Doctoral Researcher | Process Control & Precision Metrology

Focus & Aberration Control: Designed a wavefront sensorless closed loop system for focus control using an optofluidic spatial light modulator. Restoring image quality to diffraction limited. Published in ACS Photonics 2025.

Design of Experiments & System Optimization: Conducted systematic Design of Experiments and ZEMAX/Python tolerance analyses to optimize optical imaging microscopes. Applied iterative optimization algorithms to restore submicron resolution.

Metrology Integration: Engineered and integrated holography metrology modules into complex Research and Development platforms; managed technology transfer from ideas, prototype and to applications in biology.

Lab Operations: 4 years managing a high power ultrafast laser and a custom two-photon microscope: Environmental Health and Safety (EH&S) compliance, high precision equipment calibration, and technical documentation.

Jun'20 | Dec'21 — **Leibniz Institute of Photonic Technology** — Jena, Germany
Master of Science Assistant Researcher | Spectroscopist & Data Analytics

Optomechanical Design: Built a compact 1064 nm fiber probe Raman spectrometer. Optimized Near Infrared excitation for sensitive detection and background reduction in fluorescent samples.

Statistical Process Control | SpectraMap: Developed a Python package for automated classification on high dimensional hyperspectral maps. Published in Applied Sciences 2024.

Mar'18 | Mar'19 — **INTECOL S.A.S.** — Medellín, Colombia
Junior Machine Vision Engineer | Automated Optical Inspection Specialist

Automated Optical Inspection: Developed AI driven machine vision algorithms for high speed quality control. Engineered Halcon inspection pipelines and Root

Cause Analysis methodologies to detect labels and micro defects on production lines.

Technical Competencies

Systems Engineering & Metrology: Wavefront sensorless closed loop control based system development (*ACS Photonics 2025*). Using holography for phase measurements and iterative optimization algorithms for aberration correction.

Imaging Systems & Process Control: Knowledge of high resolution imaging systems via confocal and two-photon microscopy. Also training from ASML and ZEISS SMT training.

Data Analytics & Statistical Process Control: Built SpectraMap for automated classification and Statistical Process Control on high dimensional maps (*Applied Sciences 2024*). Proficiency in NumPy, SciPy and scikit learn.

Verification & Validation: Development of custom Automated Optical Inspection pipelines for manufacturing; Expertise in Halcon, C++, MATLAB and ZEMAX optimization.

Education

Jan'22 | Jun'26 — **Medical University of Innsbruck**
Austria — *Doctor of Philosophy in Adaptive Optics & Precision Metrology*

2019 | 2021 — **Friedrich Schiller Univ. Jena**
Germany — *Master of Science in Photonics*

2012 | 2018 — **University of Cauca**
Colombia — *Bachelor of Science in Physics Engineering*

Publications & Awards

Awards: Selected for **Advanced Semiconductor Materials Lithography (ASML)** and **ZEISS Semiconductor Manufacturing Technology (SMT)** (optical imaging systems). COLFUTURO 2026.

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills & Hobbies

Project Ownership: developer of microscopy system from optical design to publication (*ACS Photonics 2025*); independently built the SpectraMap Python package (*Applied Sciences 2024*).

Interdisciplinary Teamwork: Collaborated with clinical and hardware teams across INTECOL S.A.S., Leibniz Institute of Photonic Technology (IPHT) Jena, and Medical University of Innsbruck.

Systematic Problem Solving: Diagnosed and resolved systematic production defects at INTECOL.

Hobbies: Mountain biking, Bachata, Harmonica.

Languages

English (Fluent) · German (Upper Intermediate B2) · Spanish (Native)

References

Prof. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft
Leibniz-IPHT Jena
christoph.kraft@leibniz-ipht.de

Leibniz ipht
LEIBNIZ-INSTITUT *für* PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020

F. Sondermann Juan D. M.

Prof. J. Popp	F. Sondermann	Juan David Munoz Bolanos
Chairman	Executive Member	Guest
of Leibniz-IPHT	of Leibniz-IPHT	

Leibniz | ipht
LEIBNIZ-INSTITUT *für* PHOTONISCHE TECHNOLOGIEN

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Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

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Jena, 13 November 2020

F. Sondermann

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20 DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC OF COLOMBIA.

UNIVERSIDAD DEL CAUCA

ON BEHALF OF THE

REPUBLIC OF COLOMBIA

AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION

CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS

CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS, GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR E INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINJUSTICIA 1997

OFFICIAL TRANSLATION No. 0439
20 DECEMBER 2018

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich-Schiller-Universität Jena
2

Prof. Dr. Christian Spielmann
Dekan

Prof. Dr. Silvana Botti
Vorsitzende des Prüfungsausschusses

FRIEDRICH-SCHILLER-
UNIVERSITÄT
JENA

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz • deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen 1998

ÖSD ist Mitglied von

ALTE Association of Language Testers in Europe

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Sprachinitiativen 1998

ÖSD ist Mitglied von

ALTE Association of Language Testers in Europe

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25m40434

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Dr. Thomas Marschner
Program Manager Funding Projects

Dr. Thomas Stammler
CTO & Head of Product Strategy

ZEISS

ASML

PhD Master Class

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

PhD Master Class

Thursday, March 19

19:30 – 22:30	Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

PASAPORTE
PASSPORT

REPUBLICA DE COLOMBIA

Tipo / *Type*
P

Cod. país / *Country code*
COL

Pasaporte Nº / *Passport No.*
AV410884

Apellidos / *Surname*
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Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*
04 JUN/JUN 1994

Núm. personal / *Personal No.*
CC1061772278

ORD AV410884 10 DIC/DEC 2028

Sexo / *Sex*
M

Lugar de nacimiento / *Place of birth*
LA CRUZ COL

Fecha de expedición / *Date of issue*
11 DIC/DEC 2018

Autoridad / *Authority*
G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*
10 DIC/DEC 2028

Firma del titular / *Holder's signature*
Juan David M

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